Development Of Bone , Cartilage and Joints

Learning objectives

- Discuss Histogenesis of Bone
- Know the Intramembranous Ossification
- Describe the Endochondral Ossification
- Mention the Ossification of limb bones
- Describe the development of joints
- Describe the development of cartilage

DEVELOPMENT OF BONE

HISTOGENESIS

Bone formation during embryonic development may occur in two ways:

a) Intra membranous ossification

b) Intra cartilaginous ossification

- The bone that is formed is identical histologically.
- The first bone formed is primary bone which is latter reabsorbed and replaced by secondary bone.
- Secondary bone continues to be reabsorbed throughout life.

INTRAMEMBRANOUS BONE FORMATION

- Most flat bones are formed by Intramembranous ossification.
- The process occurs in a richly vascularised mesenchymal tissue, whose cells contact each other via long processes.
- Mesenchymal cells differentiate into osteoblasts that secrete bone matrix, forming a network of spicules & trabeculae whose surfaces are populated by these cells.

- This region of initial osteogenesis is known as <u>Primary centre of ossification.</u>

- The collagen fibers of these developing spicules & trabeculae are randomly oriented as in primary bone.

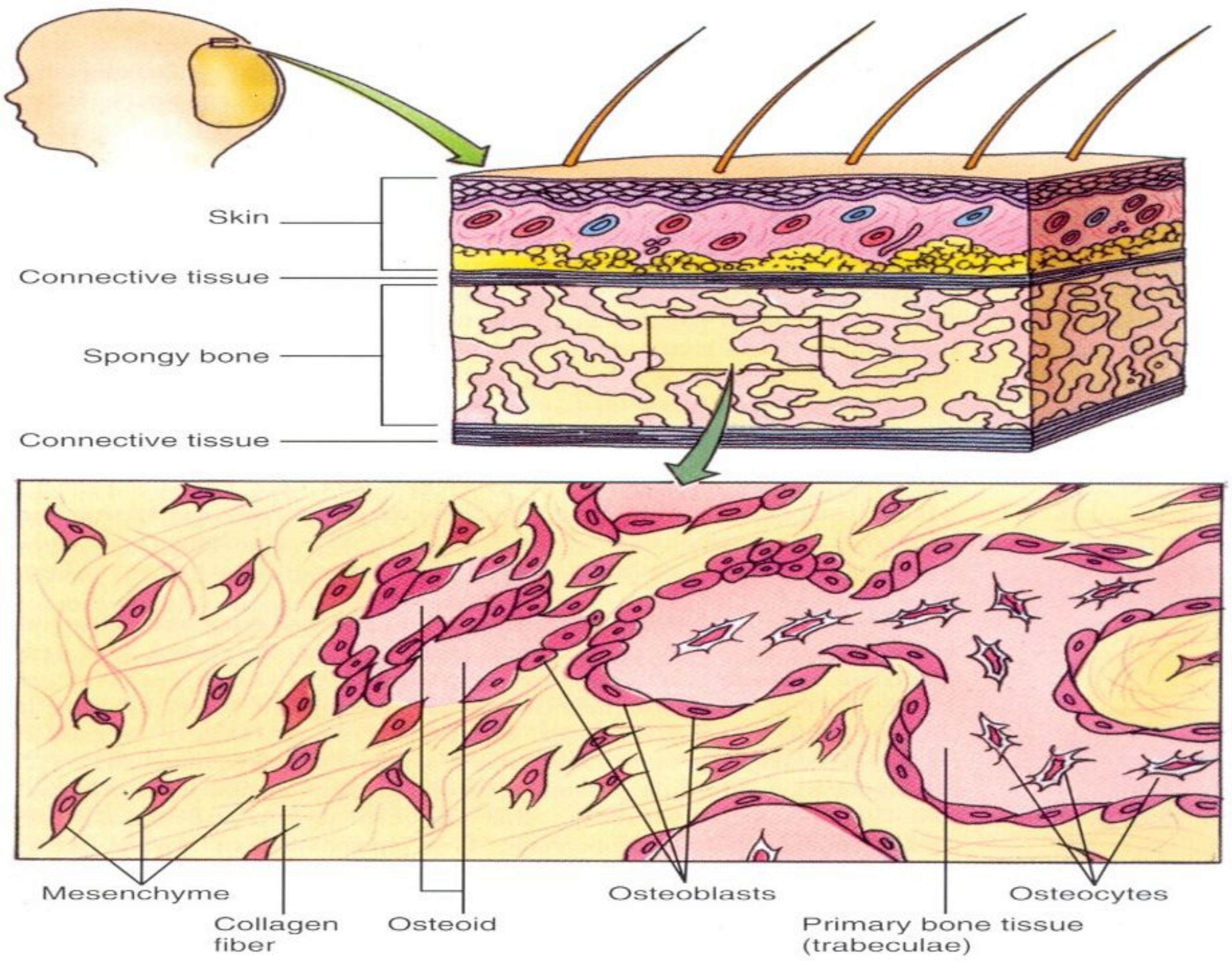
- Calcification quickly follow osteoid formation and osteoblasts trapped in their matrix become osteocytes.

- The processes of these osteocytes are also surrounded by forming bone, establishing a system of canaliculi.

- Continuous mitotic activity of mesenchymal cells provide a supply of undifferentiated osteogenic progenitor cells which form osteoblasts.

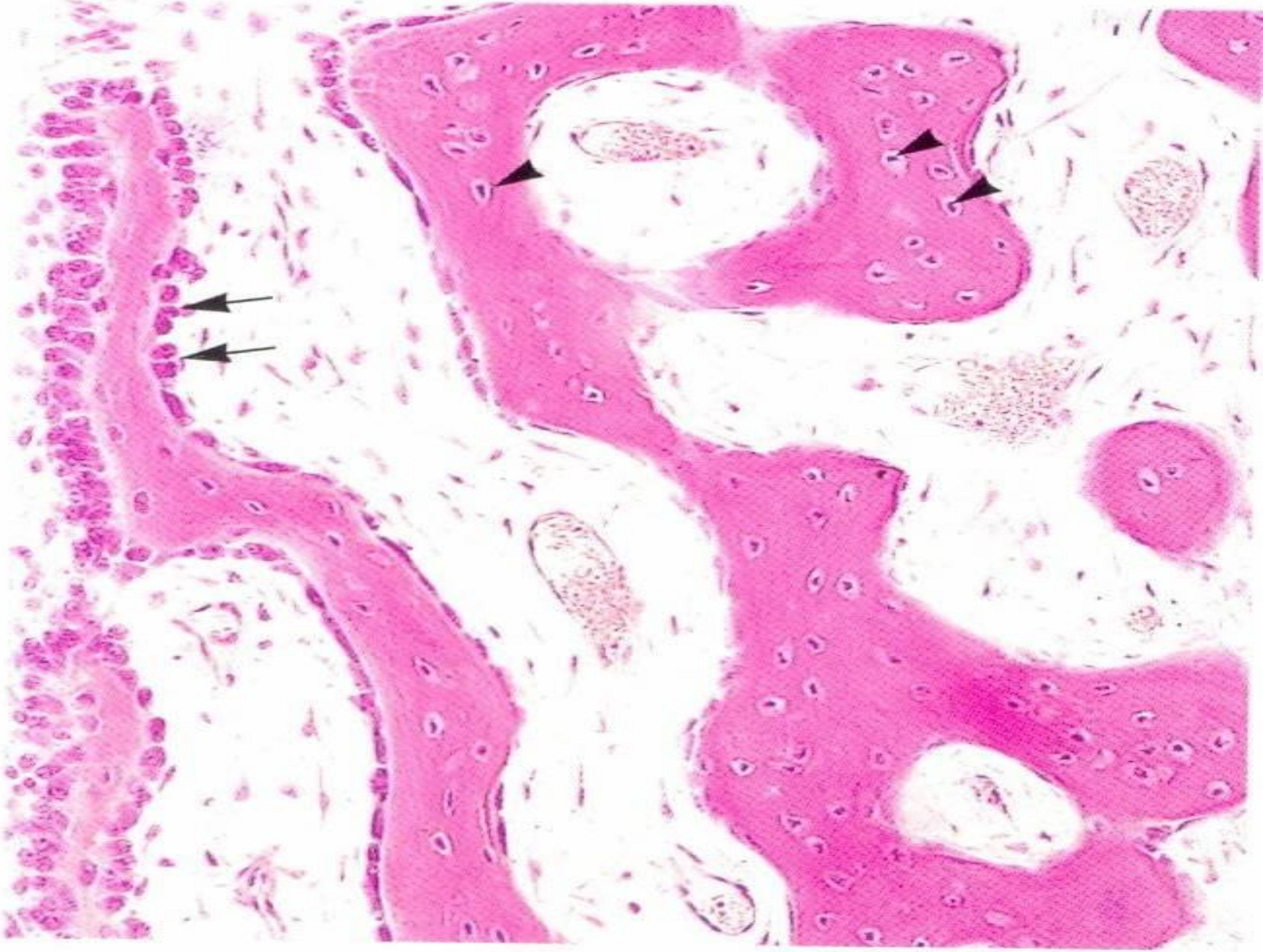
- The sponge-like network of trabeculae is established into bone marrow.
- The addition of trabeculae in periphery increases the size of bone.

- The fontanelles (soft spots) on the frontal and parietal bone of new born infants represent ossification centers that are not fused prenatally.
- Regions of mesenchymal cells that remain uncalcified represent periosteum & endosteum.
- The spongy bone deep to periosteum & periosteal layer of duramater of flat bones are transformed into compact bone forming inner and outer tables with intervening diploe.



Skin
Connective tissue
Spongy bone
Connective tissue
Mesenchyme
Collagen fiber
Osteoid
Osteoblasts
Primary bone tissue (trabeculae)
Osteocytes

F
b



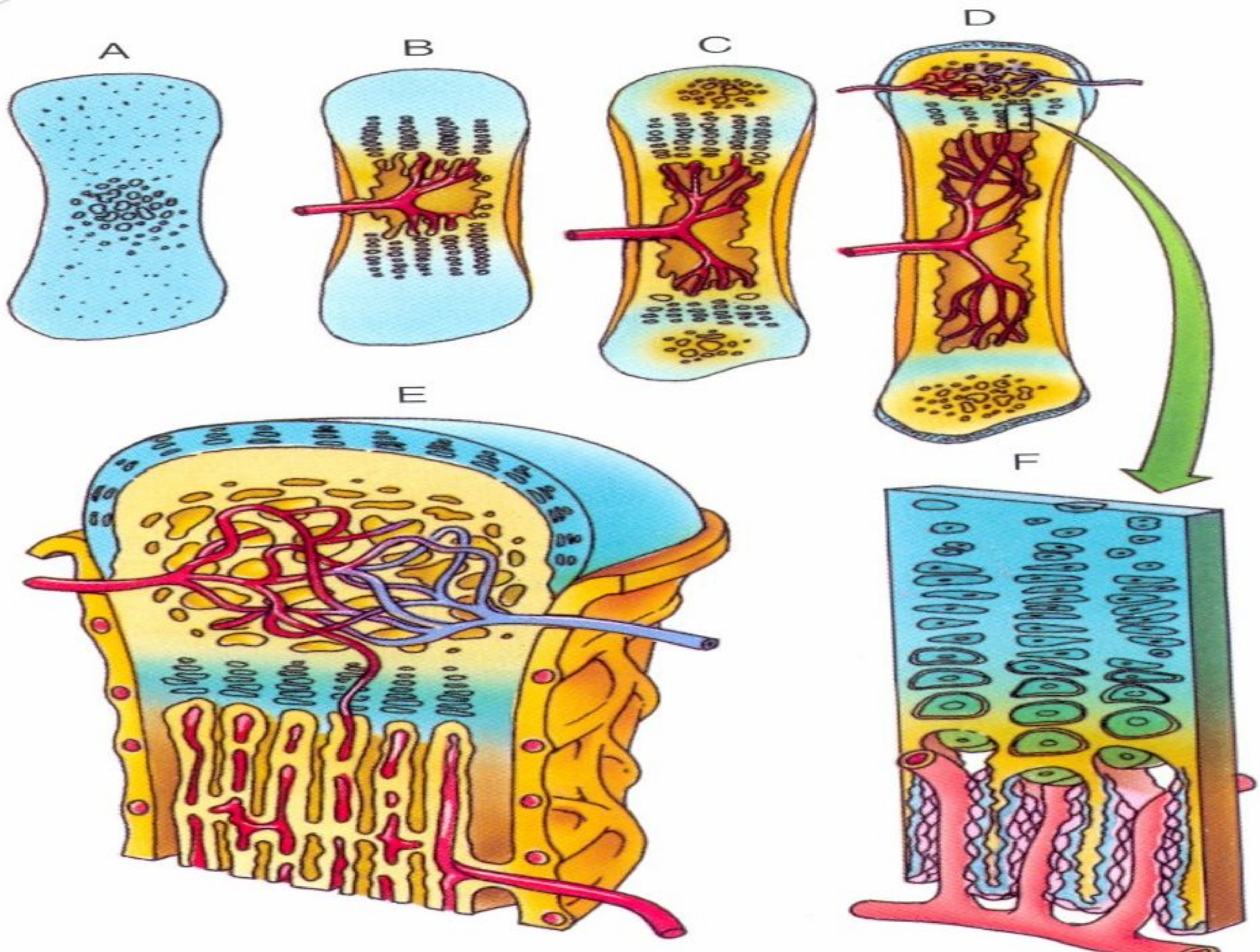
EVENTS IN ENDOCHONDRAL BONE FORMATION

- In a region where bone has to grow within the embryo, a hyaline cartilage model of that bone is developed.
- This event begins in exactly the same manner as that of hyaline cartilage developing at any location.
- The chondrocytes, in the centre of cartilage model hypertrophy, accumulate glycogen in their cytoplasm and become vacuolated.

- **Hypertrophy of chondrocytes results in enlargement of their lacunae and reduction in the intervening cartilage matrix septa, which become calcified .**

ENDOCHONDRAL BONE FORMATION

- Most of long and short bones of body develop by endochondral bone formation.
- This type of bone formation occurs in two steps:
  1) Miniature hyaline cartilage model is formed.
  2) Cartilage model continue to grow and serve as structured scaffold for bone development is reabsorbed and replaced by bone.


A
B
C
D
E
F

PRIMARY CENTRE OF OSSIFICATION

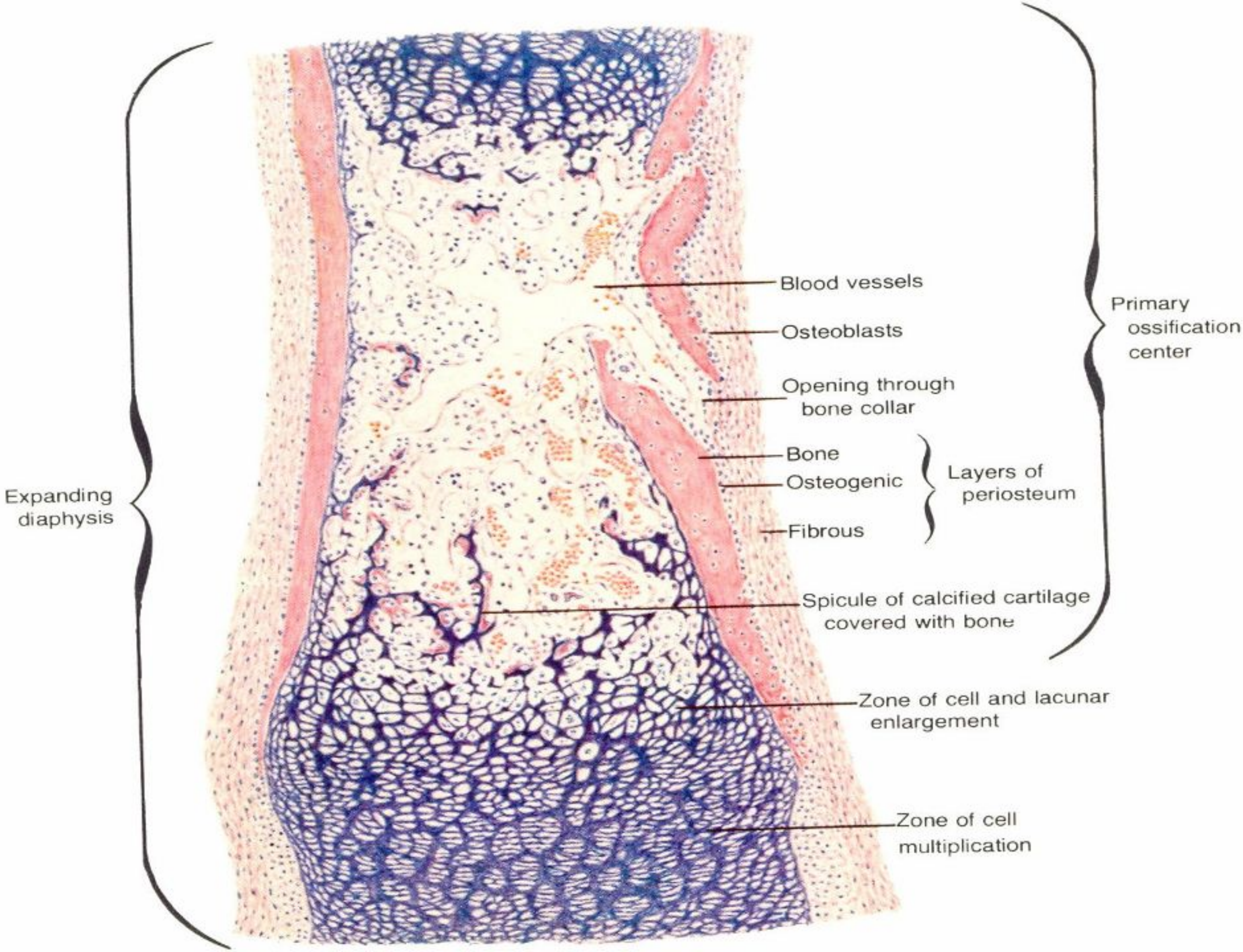
- Concurrently the Perichondrium at the middle of diaphysis of cartilage becomes vascularised.
- Chondrogenic cells becomes osteoproginator cells forming osteoblasts and occur in overlying perichondrium which becomes periosteum.
- The newly formed Osteoblasts secrete bone matrix forming the periosteal bone collar on the surface of cartilage template by intra membranous bone formation.

- The bone collar prevents the diffusion of nutrients to the hypertrophied chondrocytes within the core of cartilage model, causing them to die.
- This process is responsible for presence of empty, confluent lacunar forming large concavities the future marrow cavity in the centre of cartilage model.
- Holes in bone collar by osteoblasts permit an osteogenic or periosteal bud, composed of osteogenic progenitor cells, haemopoetic cells, and blood vessels to enter the concavities within the cartilage model.

- Osteogenic progenitor cell divide to form osteoblast.
- These newly formed cells elaborate bone matrix on the surface of calcified cartilage.
- The bone matrix becomes calcified to form calcified cartilage/ calcified bone complex.
- The calcified cartilage stains basophilic where as calcified bone stains acidophilic.

- As the Subperiosteal bone becomes thicker & grows in each direction from the middle of diaphysis towards the epiphysis, osteoblasts begins reabsorbing the calcified cartilage/ calcified bone complex, enlarging marrow cavity is formed.

- As this process continues, the cartilage of diaphysis is replaced by bone except for the epiphyseal plates which are responsible for the continued growth of bone for 18-20years.



Blood vessels
Osteoblasts
Opening through bone collar
Bone
Osteogenic
Fibrous
Layers of periosteum
Primary ossification center
Expanding diaphysis
Spicule of calcified cartilage covered with bone
Zone of cell and lacunar enlargement
Zone of cell multiplication

BONE GROWTH IN LENGTH

- The continued lengthening of bone depends on the Epiphyseal plates, whose chondrocytes proliferate and participate in the process of endochondral bone formation.
- Proliferation occurs at the Epiphyseal aspect, and replacement by bone takes place at the diaphyseal side of the plate.
- Histologically, the Epiphyseal plate is divided into five recognizable zones.
- These zones, beginning at the Epiphyseal side, are:

Five Recognizable Zones Of The Epiphyseal Plate :

1.Zone of Reserve cartilage:

- Chondrocytes randomly distributed throughout the matrix are mitotically active.

2.Zone of Proliferation:

- Chondrocytes, rapidly proliferating, form rows of isogenous cells that parallel the direction of bone growth.

3.Zone of maturation and hypertrophy:

- Chondrocytes mature, hypertrophy, and accumulate glycogen in their cytoplasm.
- The matrix between their lacunae narrows with a corresponding growth of lacunae

4.Zone of Calcification:

- Lacunae become confluent & hypertrophied.
- Chondrocytes die, and cartilage matrix becomes calcified.

5.Zone of Ossification:

- Osteoprogenitor cells invade the area and differentiate into Oestoblasts, which elaborate matrix than becomes calcified on the surface of calcified cartilage.
- This is followed by resorption of the calcified cartilage/ calcified bone complex.

- **As long as the rate of mitotic activity in the zone of proliferation equals the rate of resorption in the zone of ossification, the Epiphyseal plate remains the same width, and the bone continues to grow longer.**

- **At about 20th year of age, the rate of mitosis decreases in the zone of proliferation, and the zone of Ossification overtakes the zones of proliferation and cartilage reserve.**

- The cartilage of the Epiphyseal plate becomes replaced by a plate of calcified cartilage/ calcified bone complex, which becomes resorbed by Osteoclastic activity, and the marrow cavity of the epiphysis.

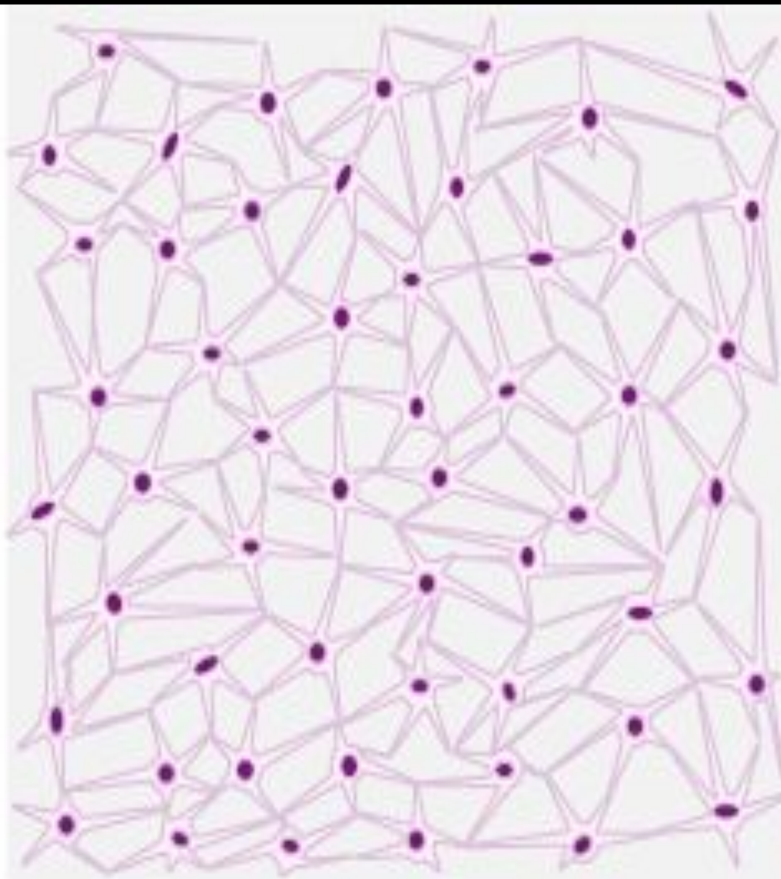
- The Epiphyseal plate is resorbed, growth in length is no longer possible.

Cartilage

It is specialized type of connective tissue, the firm consistency of the extra cellular matrix allow the tissue to bear mechanical stresses without permanent distortion.

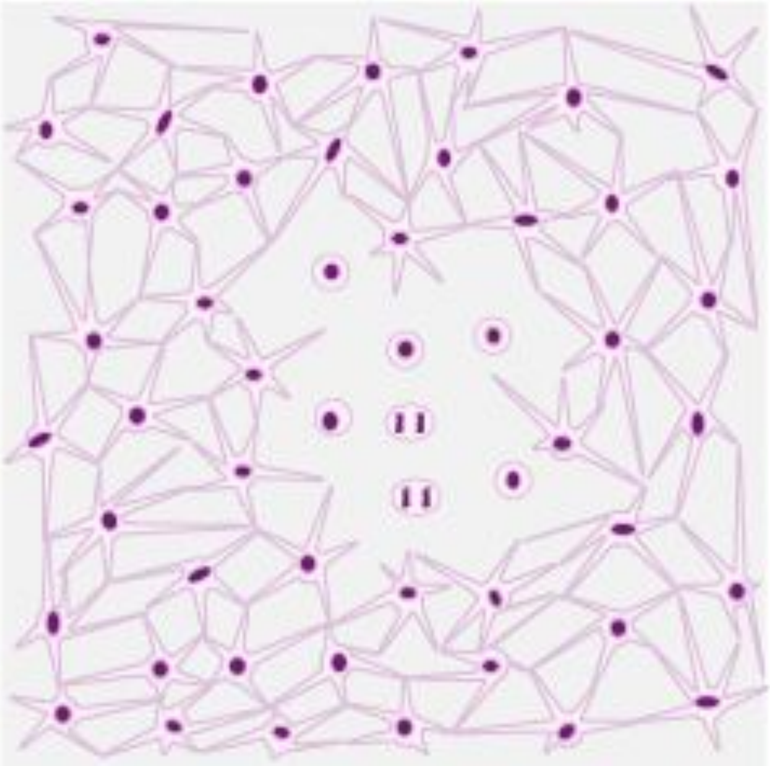
CARTILAGE DEVELOPMENT A

Avascular mesenchymal membranes consist of stem cells, imbedded in a delicate collagen matrix, that make contact with each other via cytoplasmic processes



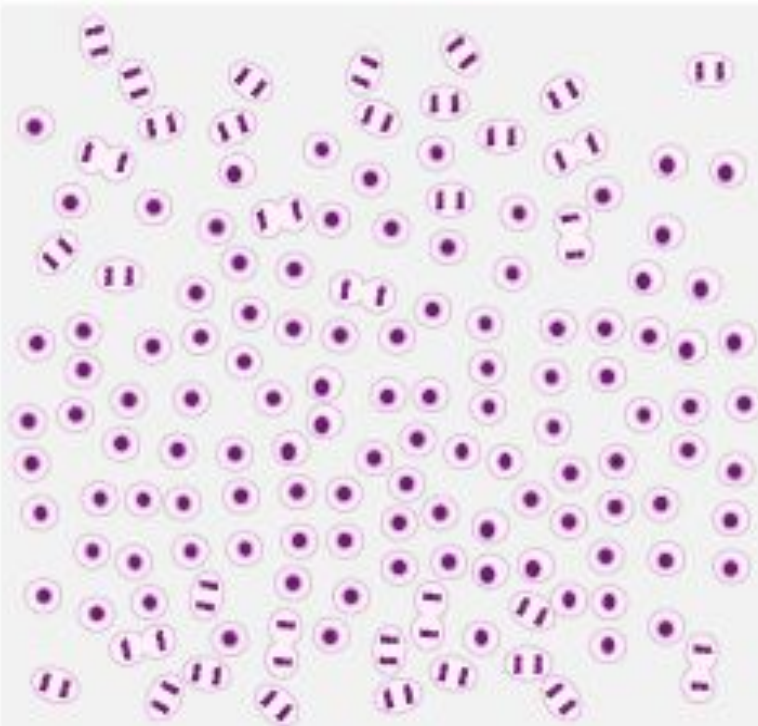
CARTILAGE DEVELOPMENT B

Mesenchymal cells begin to differentiate into protochondral cells & divide to produce chondroblasts



CARTILAGE DEVELOPMENT C

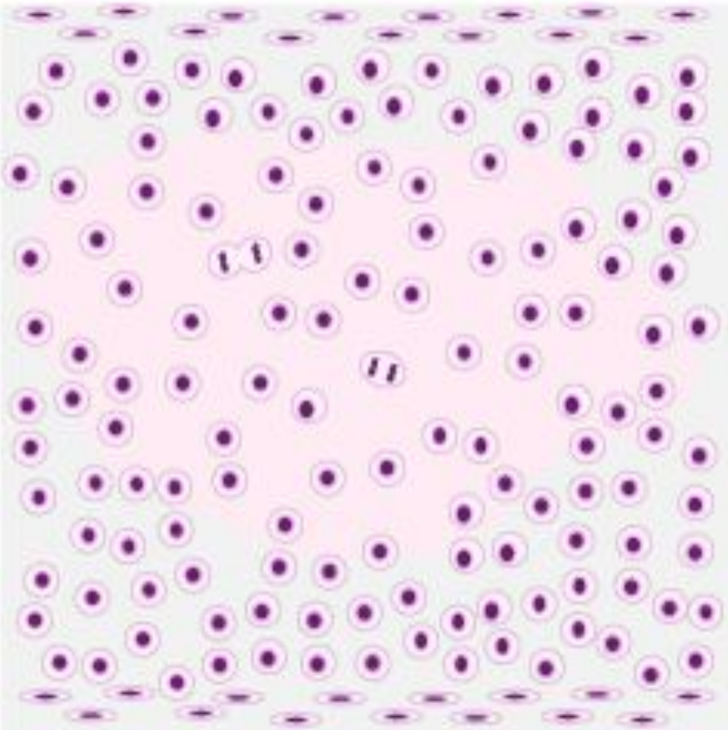
The center fills with chondroblasts as the top & bottom edges continue to differentiate



CARTILAGE DEVELOPMENT D

Stem cells remain on the edges

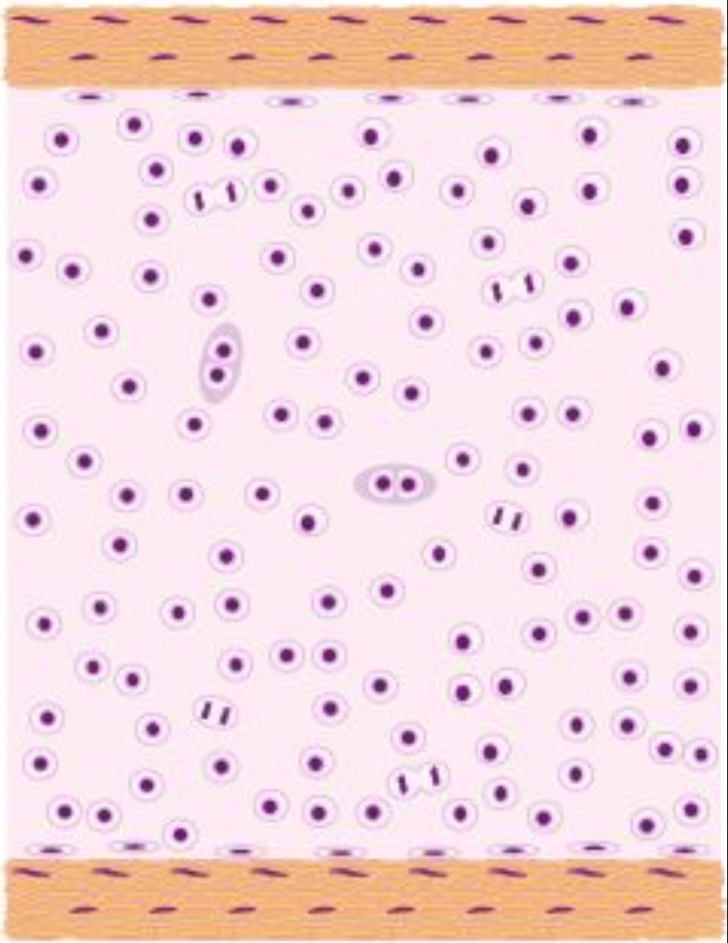
Chondroblasts in the center produce cartilage matrix & continue to divide so that the cartilage grows by interstitial growth



CARTILAGE DEVELOPMENT E

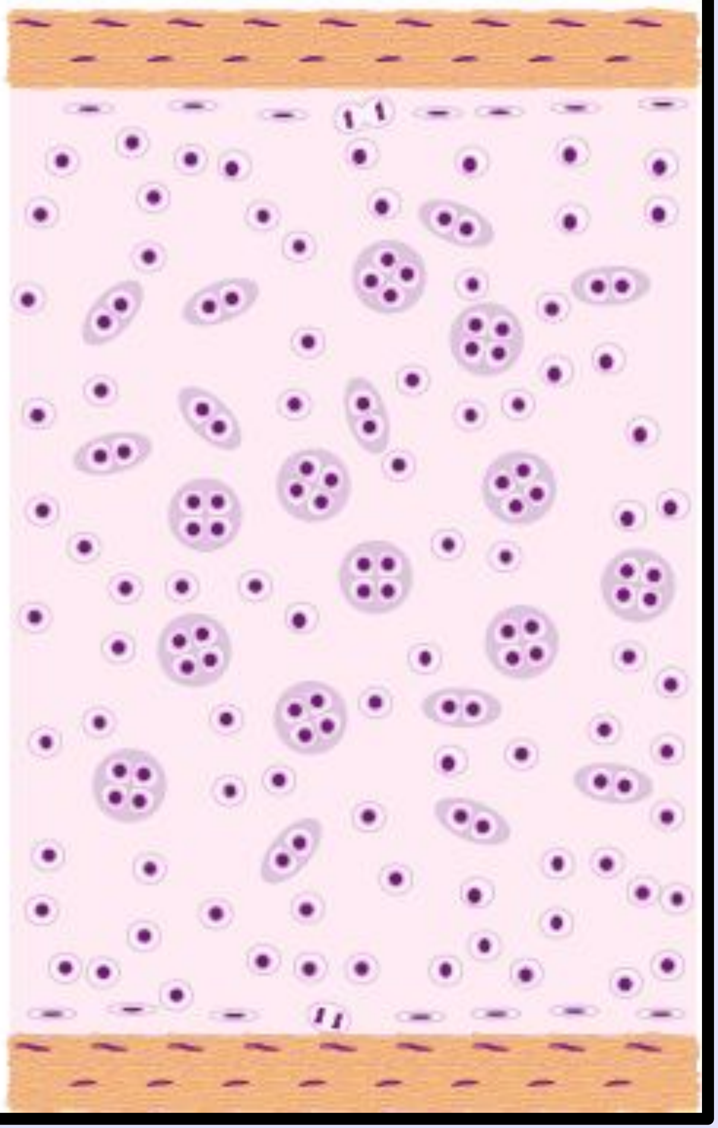
Stem cells at the edges produce fibroblasts that make the fibrous perichondrium, & add new chondroblasts to the edge by appositional growth

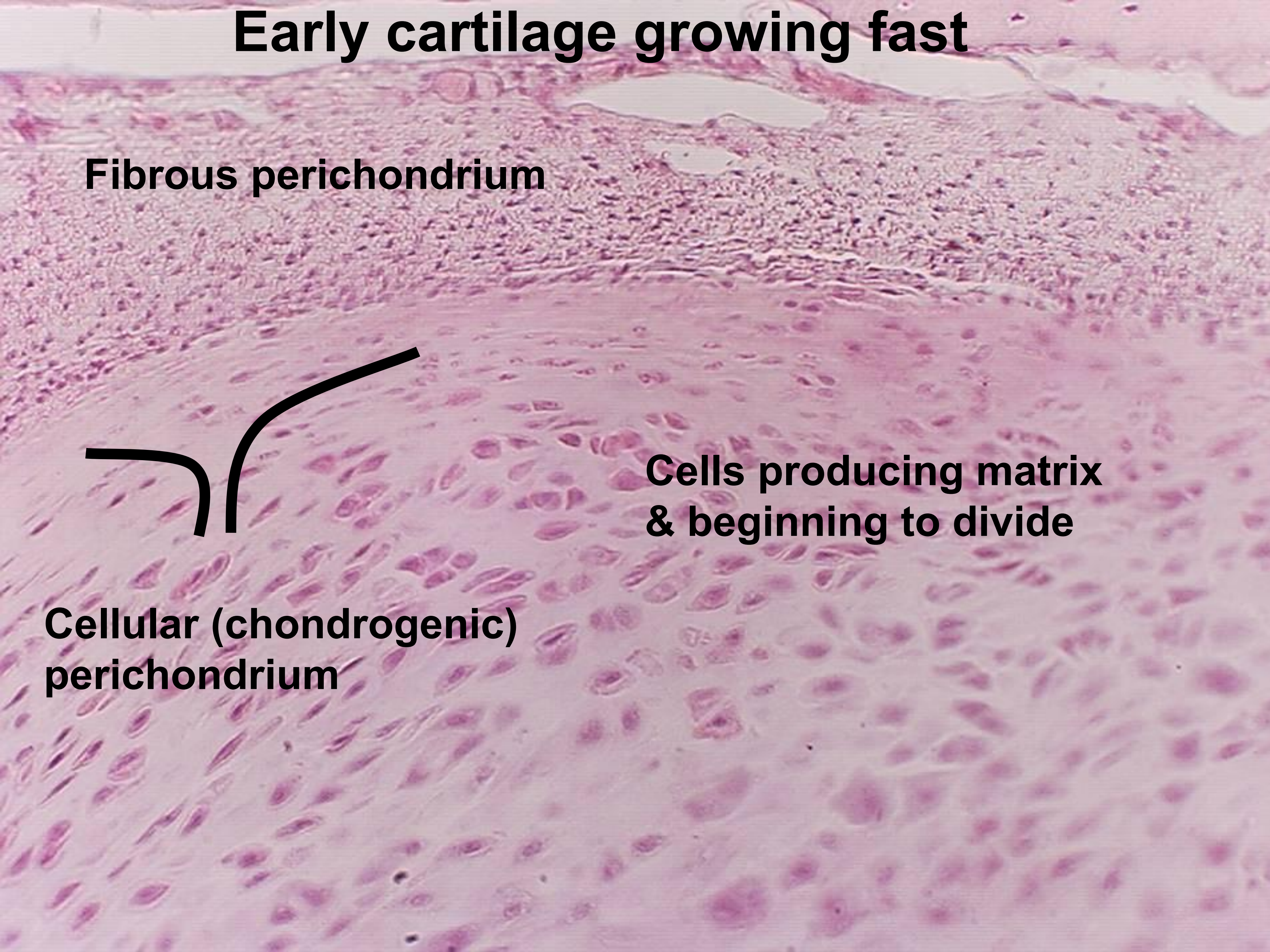
Chondroblasts in the center continue to divide, but as the matrix solidifies they don't separate



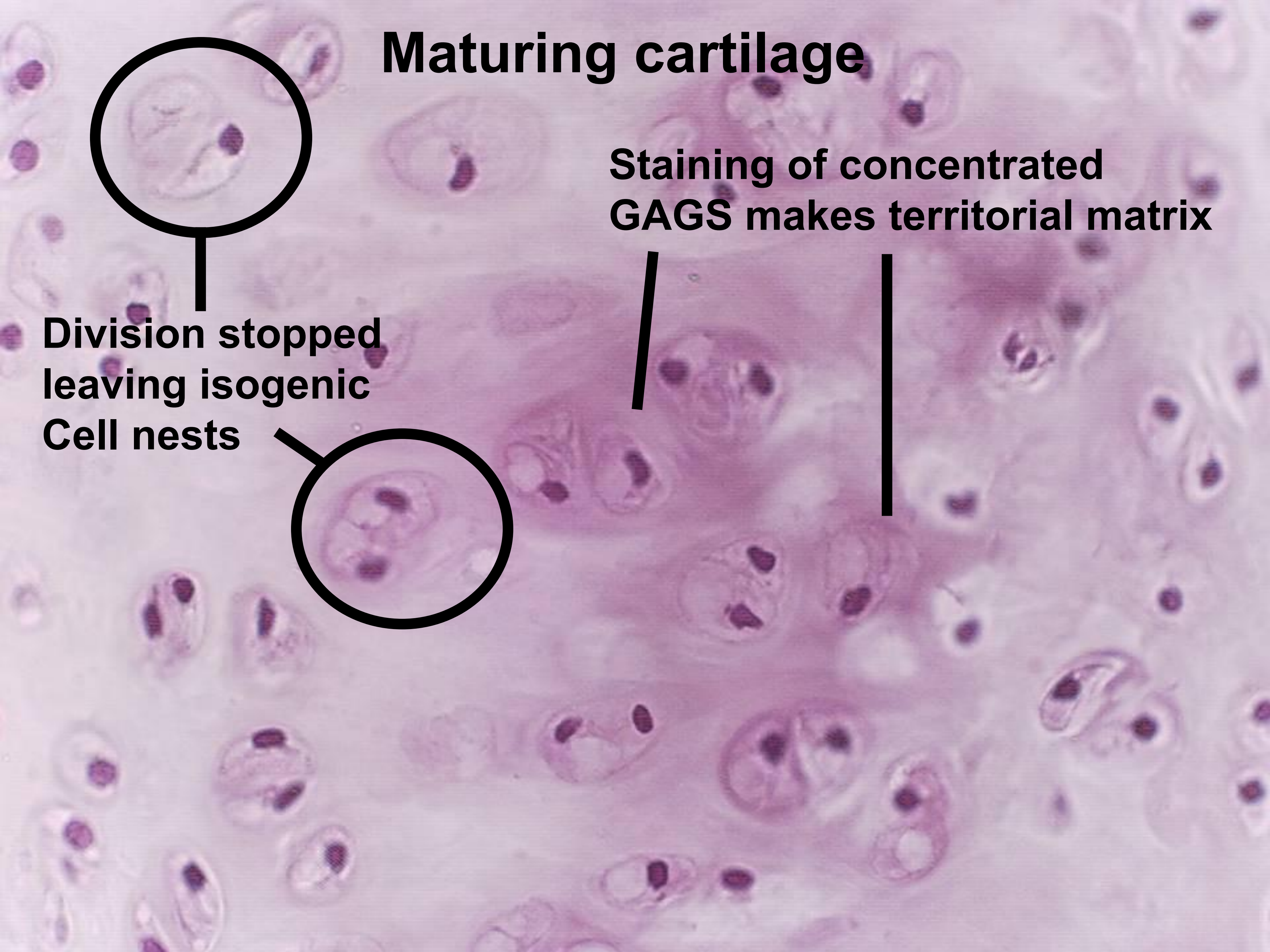
CARTILAGE DEVELOPMENT F

As the cartilage matures, stem cells in the chondrogenic perichondrium continue to generate new chondroblasts slowly by appositional growth, & chondrocytes in the center become relatively dormant



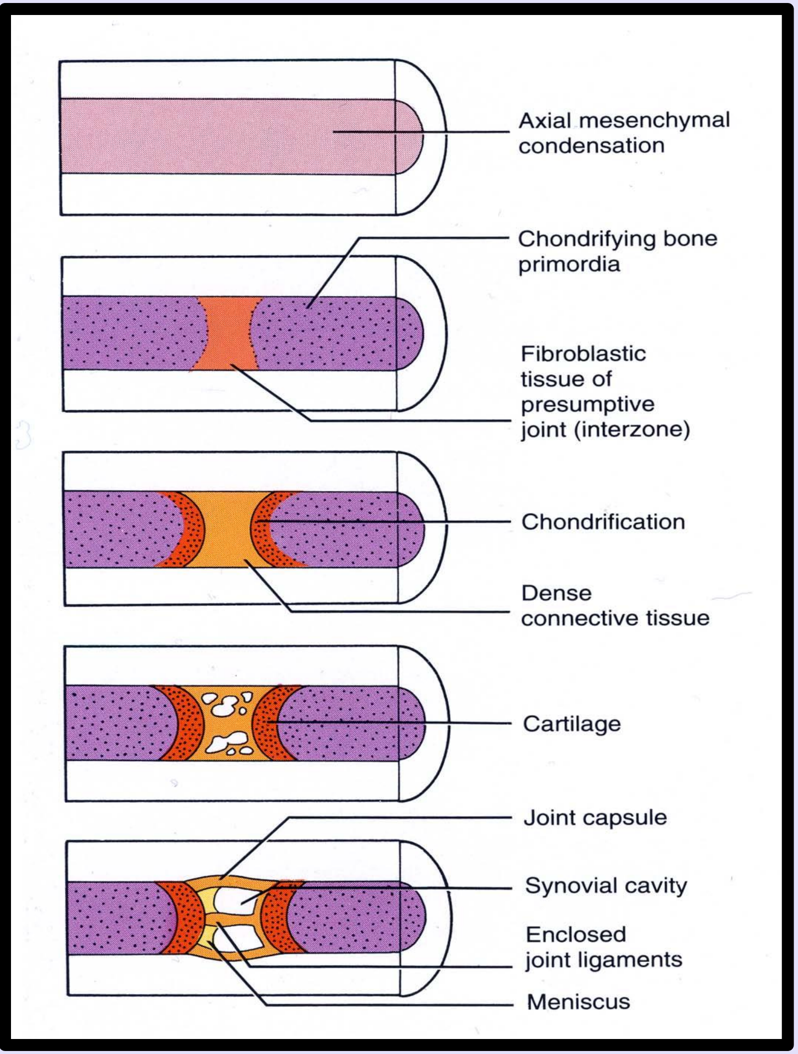

Early cartilage growing fast
Fibrous perichondrium
Cells producing matrix
& beginning to divide
Cellular (chondrogenic)
perichondrium

Maturing cartilage



Development of Joints

Joints begin to develop with the appearance of the **interzonal mesenchyme** during the sixth week, and by the end of the eighth week, they resemble adult joints.



Types of Joints

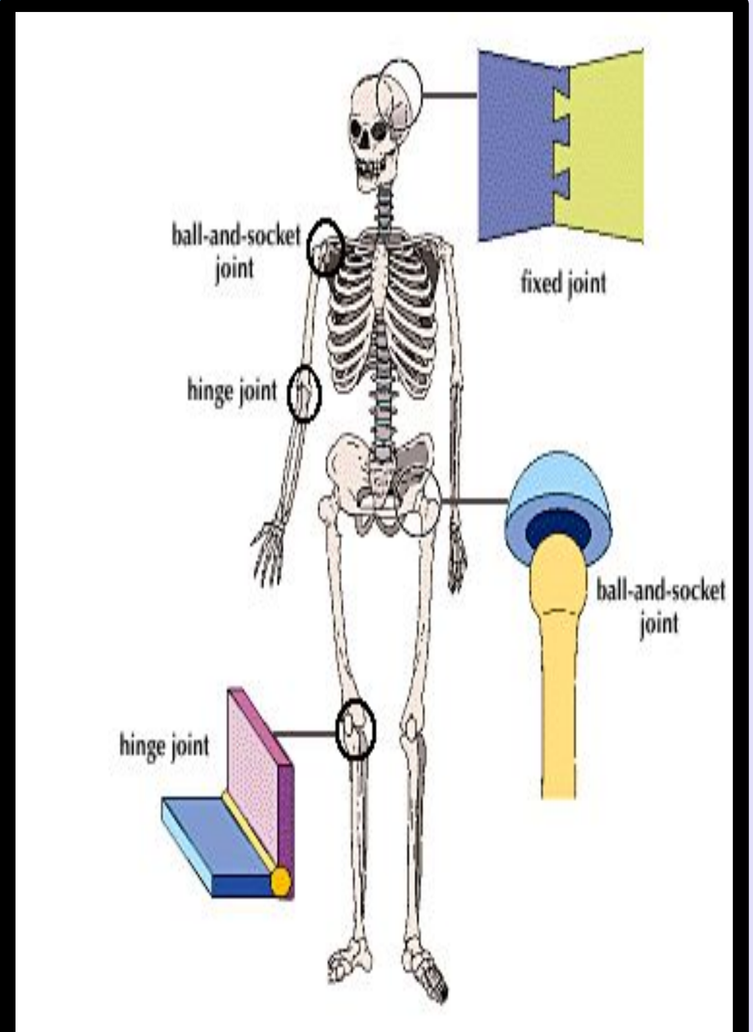
Joints are classified as

fibrous joints

Cartilaginous joints

Synovial joints. Joints

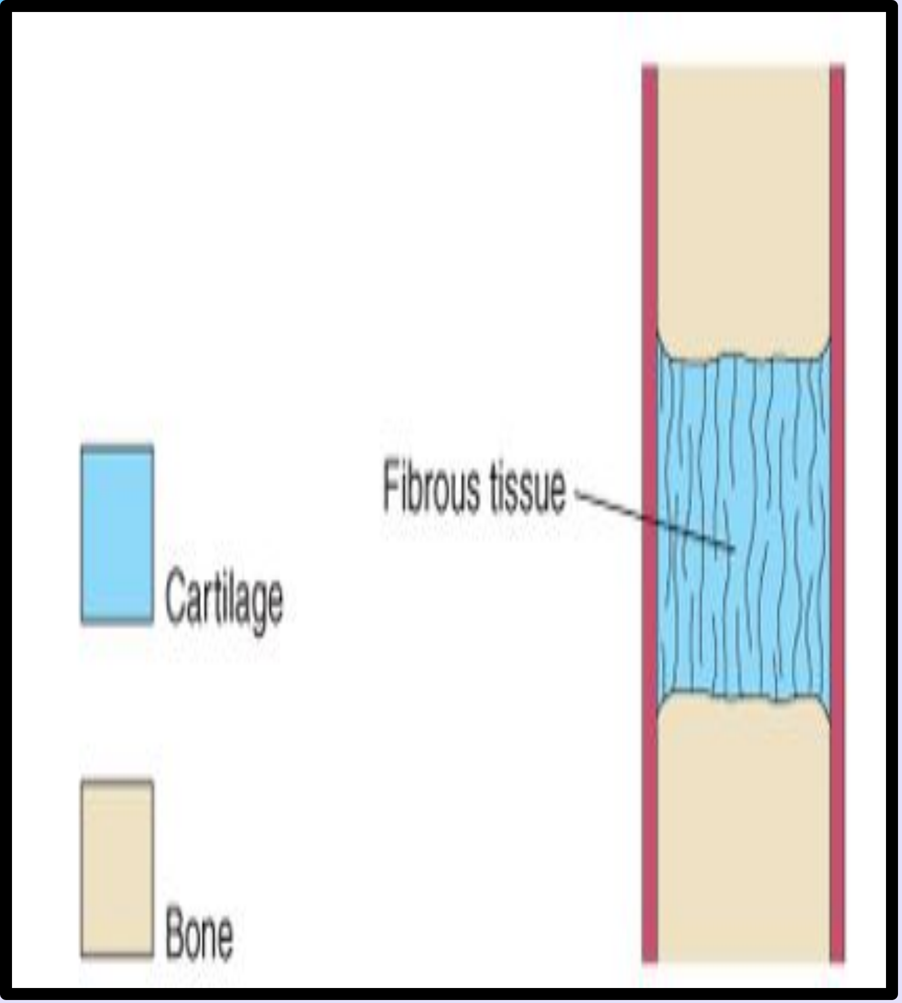
This classification of joints depends upon certain criteria



Fibrous Joints

During the development of fibrous joints, the **interzonal mesenchyme** between the developing bones differentiates into dense fibrous tissue for example, the sutures of the cranium are fibrous joints.

Bones are developed from preformed model of mesenchymal tissue which usually composed of Hyaline catilage

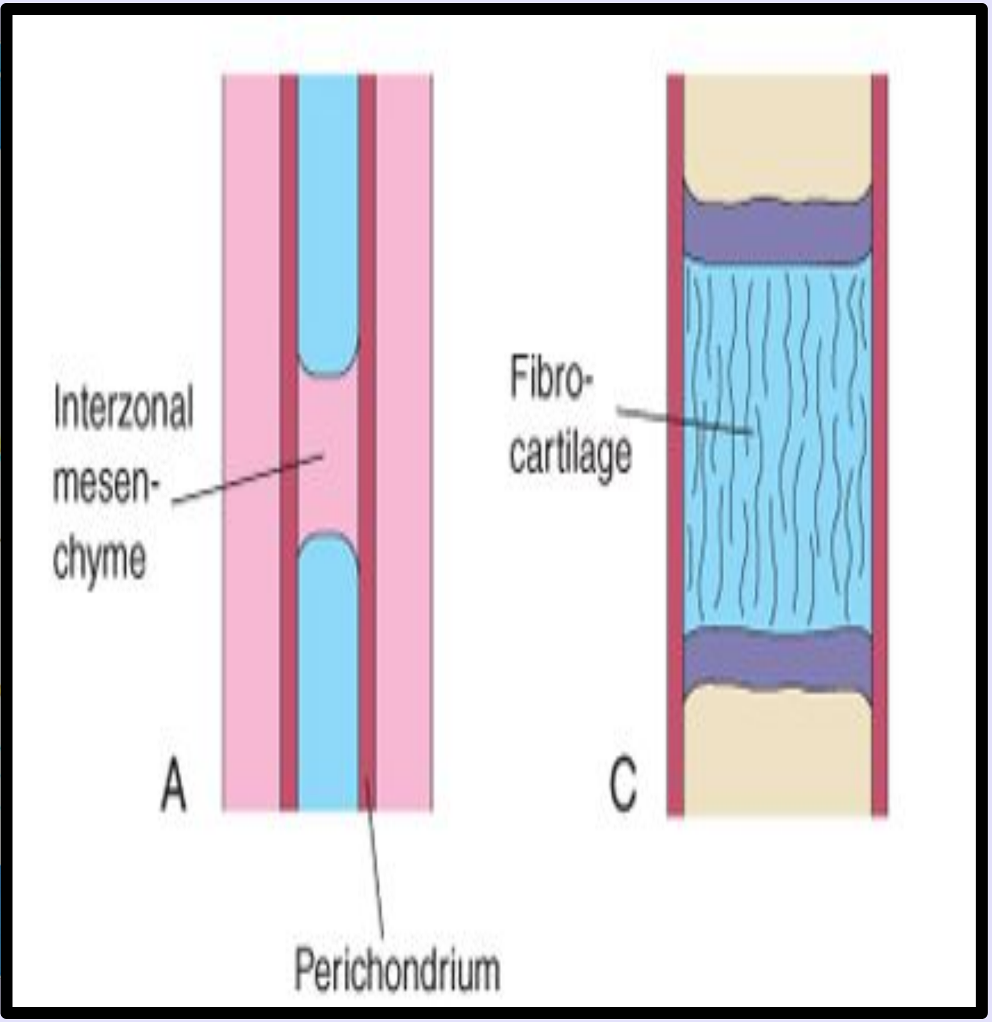


Cartilaginous Joints

During the development of cartilaginous joints, the interzonal mesenchyme between the developing bones differentiates into

Hyaline cartilage (e.g., the costochondral joints) or

Fibrocartilage (e.g., the pubic symphysis).



Synovial Joints

During the development of synovial joints (e.g., the knee joint), the interzonal mesenchyme between the developing bones differentiates as:

Peripherally it forms the capsular and other ligaments.

Centrally it disappears, and the resulting space becomes the **joint cavity** or synovial cavity.

Where it lines the joint capsule and articular surfaces, it forms the **synovial membrane** (which secretes synovial fluid), a part of the joint capsule (fibrous capsule lined with synovial membrane